Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

THIRD SEMESTER B.TECH. (Data Science)

MAKEUP EXAMINATIONS, JANUARY 2025

MATHEMATICAL FOUNDATION FOR DATA SCIENCE [MAT-2139]

REVISED CREDIT SYSTEM

Time: 9:30-12:30pm **Date: 07/01/2025** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

Q.NO	Questions	Marks	CO	PO	BTL
1A.	Find the mean deviation from the median of the following distribution. Marks: 0-10 | 10-20 | 20-30 | 30-40 | 40-50 Number of Students: 5 | 8 | 15 | 16 | 6	3	1	1,2,8, 12	1,2,3,4
1B.	Find the quartile deviation of the marks of 39 students in Statistics given below. Marks: 0-5 | 5-10 | 10-15 | 15-20 | 20-25 | 25-30 Number of Students: 4 | 6 | 8 | 12 | 7 | 2	3	1	1,2,8, 12	1,2,3,4
1C.	Find the standard deviation and coefficient of variation from the following data: Wages: <10 | <20 | <30 | <40 | <50 | <60 | <70 | <80 No of workers: 12 | 18 | 35 | 42 | 50 | 45 | 20 | 8	4	1	1,2,8, 12	1,2,3,4

2A.	Using the method of least squares, find the straight line $y = a + bx$ that best fits the following data: x: 1, 2, 3, 4, 5 y: 14, 27, 40, 55, 68	***3***	***1***	***1,2,8,12***	***1,2,3,4***
2B.	Only two factories manufacture zoggles. 20% of the zoggles from factory I and 5% from factory II are defective. Factory I produce twice as many zoggles as factory II each week. (i) What is the probability that a zoggles randomly chosen from a week's production is satisfactory? (ii) If the chosen zoggles is defective, what is the probability that it came from factory I?	***3***	***2***	***1,2,8, 12***	***1,2,3,4***
2C.	Given $f(x) = \begin{cases} kx^3; 0 < X < 1 \\ 0; otherwise \end{cases}$. (i) Find k such that the f(x) is a valid pdf, ii) Find the $P\left\{\frac{1}{4} < X < \frac{3}{4}\right\}$, $P\left\{X < \frac{1}{2}\right\}$, $P\{X > 0.8\}$.	***4***	***3***	***1,2,8, 12***	***1,2,3,4***
3A.	A machine produces bolts which are $N(4, 0.09)$, where measurements are in mm. Bolts are measured accurately and any which are smaller than 3.5 mm or bigger than 4.4 mm are rejected. Out of a batch of 500 bolts how many would be acceptable?	***3***	***2***	***1,2,8, 12***	***1,2,3,4***
3B.	Eggs laid by a particular chicken are known to have lengths normally distributed with mean 6 cm and standard deviation 1.4cm. What is the probability of (a) finding an egg bigger than 8 cm in length; (b) finding an egg smaller than 5 cm in length?	***3***	***2***	***1,2,8, 12***	***1,2,3,4***
3C.	A two-dimensional random variable (X, Y) is uniformly distributed in the region bounded by the circle $x^2 + y^2 = a^2$, Find Correlation between X and Y.	***4***	***3***	***1,2,8, 12***	***1,2,3,4***
4A.	Let $\bar{X}$ be the mean of the random sample of size n from a normal distribution with mean μ and variance 100. Find n so that $\Pr\{\mu - 5 < \bar{X} < \mu + 5\} = 0.954$.	***3***	***3***	***1,2,8, 12***	***1,2,3,4***
4B.	If X has pdf $f(x) = \lambda e^{-\lambda(x-a)}$if $x \geq a$. Find its moment generating function and hence find the mean and variance.	***3***	***4***	***1,2,8, 12***	***1,2,3,4***
4C.	The joint probability mass function of random variable (X, Y) is given by $p(x, y) = k(2x + 3y)$; $x = 0, 1, 2$ and $y = 1, 2, 3$. Find the (a) marginal probability distributions of X and that of Y, (b) conditional distribution of Y given X= 2.	***4***	***4***	***1,2,8, 12***	***1,2,3,4***

<table>
<tr>
<td>5A.</td>
<td>A drug is given to 8 patients and differences in their BP were recorded to before drug A: 112, 113, 118, 120, 119, 113, 110, 122 and after drug A: 116, 120, 117, 125, 126, 111, 111, 117. Is it reasonable to believe that the drug has no effect on change of BP? Test the significance with 2.365 at 5% Level of Significance.</td>
<td>3</td>
<td>5</td>
<td>1,2,8, 12</td>
<td>1,2,3,4</td>
</tr>
<tr>
<td>5B.</td>
<td>A survey of 320 families with 5 children each revealed the following distribution
<table>
<tr><td>No. of boys</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr>
<tr><td>No. of girls</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr>
<tr><td>No. of families</td><td>14</td><td>56</td><td>110</td><td>88</td><td>40</td><td>12</td></tr>
</table>
Is this result consistent with the hypothesis that male and female births are equally probable? Test the hypothesis at 5% Level of Significance with 11.07?</td>
<td>3</td>
<td>5</td>
<td>1,2,8, 12</td>
<td>1,2,3,4</td>
</tr>
<tr>
<td>5C.</td>
<td>Following table gives four different crops treated with three different fertilizers. Test at 1% level of significance whether

(i) there is significant difference in yield due to fertilizers

(ii) there is significant difference in yield due to crop
<table>
<tr><td>Crop→
Fertilizer↓</td><td>1</td><td>2</td><td>3</td><td>4</td></tr>
<tr><td>1</td><td>4.5</td><td>6.4</td><td>7.2</td><td>6.7</td></tr>
<tr><td>2</td><td>8.6</td><td>7.8</td><td>9.6</td><td>7.0</td></tr>
<tr><td>3</td><td>5.9</td><td>6.8</td><td>5.7</td><td>6.7</td></tr>
</table>
Test the significant difference at 1% LOS with the tabulated value at 10.93 and 9.78.</td>
<td>4</td>
<td>5</td>
<td>1,2,8, 12</td>
<td>1,2,3,4</td>
</tr>
</table>